

RESEARCH ARTICLE



HPV vaccination coverage, hesitancy, and barriers: Insights from a serial cross-sectional study in Shanghai, China (2019–2024)

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ABSTRACT

Cervical cancer is the fourth most common cancer among women globally and a significant public health challenge in China, which accounts for nearly 18% of global cervical cancer cases. The HPV vaccine is a proven intervention for preventing high-risk HPV infections and associated cancers. This serial cross-sectional study conducted in Shanghai, China, in 2019 and 2024 analyzed changes in HPV vaccination rates and vaccine hesitancy. Multivariable logistic regression was used to assess the associations between HPV vaccination, general vaccine hesitancy, and HPV-specific vaccine hesitancy, adjusting for socio-demographic factors. The Population Attributable Fraction (PAF) quantified the proportion of non-vaccination attributable to these factors. In 2019 and 2024, two waves of data were collected from 1,037 and 1,450 parents, respectively. Results showed an increase in HPV vaccination from 6% to 30% during the study period, primarily among women aged 18 and older, while coverage among the WHO target group (girls aged 9–14) remained low at 8%. Parents with HPV-specific vaccine hesitancy were 75% less likely to vaccinate their daughters (OR = 0.25, 95% CI: 0.11–0.56), while general vaccine hesitancy reduced vaccination likelihood by 94% (OR = 0.06, 95% CI: 0.00–0.89). PAF analysis indicated that HPV-specific vaccine hesitancy contributed to 24% of non-vaccination cases, higher than general vaccine hesitancy (10%). These findings highlight the critical need for targeted interventions addressing HPV-specific barriers, improving education about vaccination timing and options, and reducing logistical obstacles to align with global cervical cancer elimination goals.

ARTICLE HISTORY

Received 14 February 2025
Revised 14 May 2025
Accepted 26 May 2025

KEYWORDS

HPV vaccine; cervical cancer prevention; vaccine hesitancy; Population Attributable Fraction (PAF); China public health

Introduction

Cervical cancer is the fourth most common cancer among women globally, with an estimated 604,000 new cases and 342,000 deaths in 2020.¹ The Human Papillomavirus (HPV) vaccine has long been proven to provide effective protection against high-risk HPV strains, including those which cause cervical and other cancers. To decrease this burden, the World Health Organization (WHO) has set a target of ensuring 90% of girls are fully vaccinated with the HPV vaccine by age 15 as part of a global effort to eliminate cervical cancer.² Despite its proven effectiveness, significant disparities persist in vaccine access and uptake. Low- and middle-income countries (LMICs) adopted the vaccine later than high-income countries with approximately 144 countries now including the HPV vaccine in their national immunization programs (NIPs). This comprises over 98% of high-income countries, but only 46% of low-income and 59% of lower-middle-income countries.³ Rapid roll-out of HPV vaccination in LMICs is essential to narrow these inequalities and reduce the global burden of HPV-related diseases.

As one of the most populous countries globally, China faces unique challenges in integrating the HPV vaccine into its NIP, including the high cost of vaccines, limited health resources, and competing health priorities.⁴ While the WHO

recommends initiating the HPV vaccine series between the ages of 9 and 12, when immune response development are more robust^{5(p13)}, and vaccination can be completed before potential exposure to HPV.⁶ However, China's current vaccination landscape does not align with this guidance. Most HPV vaccine recipients in China are adult women aged 18 to 26, primarily due to the high cost of the vaccine.⁷ By the end of 2022, the cumulative first-dose coverage for girls aged 9 to 14 was only 4%.⁸ This gap between the recommended and actual vaccination ages emphasize the need to address barriers specific to parental decision-making, as parents are often the primary decision-makers and financial supporters for child and adolescent vaccination. Improving parental awareness and willingness to vaccinate their children at the targeted ages is critical for achieving cervical cancer prevention goals.

China has made efforts to more closely comply with the WHO cervical cancer elimination initiative guidelines. In 2022, the National Health Commission announced plans for the gradual launch of free HPV vaccination programs nationwide, starting with pilot regions.⁹ By October 2024, free HPV vaccines were made available to about 40% of girls aged 13 to 14.¹⁰ Additionally, China has approved the production of domestically manufactured bivalent HPV vaccines, such as

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Cecolin and Walrinvax, to enhance vaccine availability and affordability. As these policies evolve, understanding how parental knowledge and acceptance of the HPV vaccine may shift over time is essential for evaluating current efforts and guiding future policy decisions.

Unlike general vaccine hesitancy, which broadly includes mistrust in healthcare systems, fear of side effects, and misinformation,¹¹ HPV-specific vaccine hesitancy includes barriers unique to that vaccine. These impediments involve cultural stigma associated with a sexually transmitted infection (STI), low awareness of HPV-related cancer risks, and misconceptions that vaccination promotes early sexual activity.^{12,13} Differentiating between general vaccine hesitancy and HPV-specific vaccine hesitancy is crucial for designing targeted public health interventions.¹⁴ This distinction enables tailored strategies to address the root causes of resistance, optimize resource allocation, and ultimately increase vaccination uptake to prevent cervical cancer.

To contribute to this effort, our study conducted a serial cross-sectional survey in 2019 and 2024 in Shanghai, one of the most developed, populous and cosmopolitan cities in China. The study aims to: (1) assess changes in parents' acceptance of the HPV vaccine for their children between 2019 and 2024, (2) examine the associations between HPV vaccination, general vaccine hesitancy, and HPV-specific vaccine hesitancy, and (3) quantify the proportion of non-vaccination attributable to general and HPV-specific vaccine hesitancy. These findings will provide valuable insights to inform future public health initiatives and policies aimed at improving HPV vaccination coverage and reducing the burden of cervical cancer in China.

Methods

Study design and overview

This study used a two-wave cross-sectional design to assess trends in HPV vaccination uptake, vaccine hesitancy, and parental acceptance in Shanghai, China. The first wave of data collection occurred in 2019, followed by a second wave in 2024. Both surveys collected information on general and HPV-specific vaccine hesitancy, HPV vaccination status, willingness to vaccinate, and sociodemographic characteristics. By comparing data across the two time points and applying regression models and population attributable fraction analyses, we evaluated changes over time and quantified the impact of hesitancy on HPV vaccination outcomes.

Study population

The first wave of data collection took place in Shanghai, China, between May and July 2019. The sampling method has been detailed elsewhere.¹⁵ Briefly, all districts in Shanghai, except Chongming Island – a less-populated island distant from the city – were included in the study. Forty townships were selected based on population size, as determined by the 2010 Census. Within each township, parents were randomly selected from immunization clinics, elementary schools,

middle schools, and high schools to create a comprehensive sample representing children of all ages. Parents with at least one child aged ≤ 18 years were eligible to participate.

The second wave of data collection aimed to follow up with the parents from 2019. Newly recruited parents were from the same clinics as the parents who lost to follow up. Followed-up parents may now have children older than 18 years, while newly recruited parents were required to have at least one child aged ≤ 18 years to participate.

General and HPV-specific vaccine hesitancy

General vaccine hesitancy was assessed using the Parent Attitudes About Childhood Vaccines (PACV) scale, which includes three subdomains: general attitudes, safety and efficacy, and behavior. Details of the assessment and categorization of vaccine hesitancy have been previously published.¹⁵ Briefly, responses to questions were scored as “hesitant” (scored as 2), “not sure/don’t know” (scored as 1), or “not hesitant” (scored as 0). Scores were summed and converted to a 0–100 scale, with scores < 50 categorized as “not hesitant” and ≥ 50 as “hesitant.” A similar scoring system was applied to the subdomains.

HPV-specific vaccine hesitancy was measured using the question, “Overall, how hesitant about the HPV vaccine would you consider yourself to be?” Responses on a 5-point Likert scale were dichotomized into “not hesitant” and “hesitant.”

HPV vaccination and acceptance

In 2019, HPV vaccination status was assessed with the question, “Has one of your children received the HPV vaccine?” In 2024, parents were asked about the HPV vaccination status of each of their children.

For HPV vaccine acceptance, parents were asked about their willingness to vaccinate a hypothetical child, using an age/gender matrix for daughters and sons aged 6/9, 12, or 18 years in both 2019 and 2024. Responses were recorded on a 5-point Likert scale, ranging from “not at all willing” to “very willing.” The ages were chosen to align with recommendations: 18 years (Chinese healthcare workers' preferred initiation age,⁷ 12 years (WHO recommendation,² and 6 years (an earlier age when other vaccines are typically administered, prior to sexual activity becoming a factor). In the 2024 survey, the lower age was changed to 9 years to match HPV vaccine eligibility criteria in China.

Data were also collected on parents' self-reported knowledge of HPV and the HPV vaccine, dichotomized into low (not at all, not much) and high (some, a lot). Parents' sociodemographic characteristics, including age, family composition, education, income, residency (local/non-local), *huji* (urban/rural), and district (central/suburban), were also recorded.

Statistical analysis

The 2024 sample was weighted to match the 2019 sample distribution by age, *huji* (urban/rural), residency (local/non-local), and district to reduce sampling bias and ensure

comparability between the two datasets. A subset analysis focused on parents with daughters aged 9–17 years, who are eligible for the HPV vaccine and whose parents are the primary decision-makers for vaccination.

Logistic regression models were used to evaluate the association between HPV vaccination and general or HPV-specific vaccine hesitancy, as well as the hesitancy subdomains, adjusting for sociodemographic variables and the child's age. Odds ratios (ORs) were reported, with statistical significance assessed at an alpha level of 0.05.

The vaccination rates by age and parents' willingness to vaccinate a hypothetical child by age and gender were graphically compared between 2019 and 2024.

The adjusted Population Attributable Fraction (PAF) was determined for general and HPV-specific vaccine hesitancy factors, as well as for the subdomains, to estimate the contribution of hesitancy to non-vaccination rates. PAF was calculated using the formula:

$$PAF = p \times (rr - 1) / rr$$

where p is the proportion of vaccine-hesitant parents among non-vaccinated daughters, and rr is the odds ratio comparing the likelihood of non-vaccination among hesitant and non-hesitant groups. The rr was derived from multivariable logistic regression with non-vaccination as the outcome and vaccine hesitancy as the exposure, adjusting for sociodemographic variables. The PAF confidence intervals were reported at the 97.5% level. All analyses were conducted using SAS version 9.4 (SAS Institute, Cary, NC).

Informed consent and ethical approval

All participants provided informed consent prior to participation in each wave of data collection. The study was approved by the Health Sciences and Behavioral Sciences Institutional Review Board at the University of Michigan (#HUM00241051) and the Shanghai Centers for Disease Control and Prevention Ethical Review Committee (#2019–17–1).

AI-Use disclosure

This manuscript was written by the author, who was solely responsible for the conceptualization, study design, data interpretation, and original drafting of the content. To enhance clarity and improve language quality, the author used OpenAI's ChatGPT-4o model and Grammarly for grammar checking, language refinement, and editing support. No AI tools were used to generate or interpret research findings, nor to formulate substantive scientific arguments.

Results

Demographic characteristics

In 2019, a total of 1,037 parents participated in the first wave of data collection. In 2024, 1,450 parents were approached, with 449 (31%) reporting having daughters aged 9–17 years. Among these, 613 (42%) participants were followed up from the 2019

cohort. However, due to the lack of personal identification information to link the follow-ups, the two waves of data collection were treated as serial cross-sectional studies.

Table 1 presents the demographic characteristics of the participants. The distributions were largely consistent across the two waves. Most participants (83% in both 2019 and 2024) were mothers. A majority (62%) of parents were aged 30–39, and more than half (53% in 2019 and 50% in 2024) had a college education or higher. Most families reported a relatively high household income, were local residents of Shanghai (65%) and lived in suburban districts (70%).

General vaccine hesitancy decreased over time, with 18% of participants in 2019 reporting hesitancy compared to 11% in 2024. Both the general attitudes and safety and efficacy subdomains of vaccine hesitancy showed declines, with the latter decreasing significantly from 68% to 20%. However, the behavior subdomain exhibited the opposite trend, increasing from 10% in 2019 to 27% in 2024. For the HPV vaccine specifically, 28% of participants in 2024 expressed hesitancy. Knowledge of HPV improved over time, with 51% of parents reporting high knowledge in 2019, increasing to 67% in 2024, six years after the vaccine entered the private market in China.

General and HPV vaccine hesitancy

Table 2 displays the cross-tabulation between demographic characteristics, general vaccine hesitancy (among all parents), and HPV-specific vaccine hesitancy (among parents with daughters aged 9–17) in 2024. Across all demographic groups, HPV-specific vaccine hesitancy rates were higher than general vaccine hesitancy rates. For instance, among parents aged 40 and older, 13% were hesitant to vaccines in general, while 33% were hesitant to the HPV vaccine specifically.

Appendix Table A1 further shows the association between demographic characteristics and vaccine hesitancy subdomains among all parents in 2024. Hesitancy in the behavior subdomain was the highest (37%), followed by the safety and efficacy subdomain (19%) and general attitudes (8%).

HPV vaccination and acceptance

Figures 1 and 2 present the proportions of parents willing to vaccinate their child by age and gender. A consistent trend emerged: parents expressed higher acceptance for older children and for daughters compared to sons. Notably, there was a shift in acceptance toward older ages. In 2019, 36% of parents were willing to vaccinate their 12-year-old daughter, compared to 32% in 2024. Similarly, 31% of parents were willing to vaccinate their 12-year-old son in 2019, dropping to 25% in 2024. Figure

Figure 3 illustrates the HPV vaccination rates by age among girls aged 9 years and older who were eligible for the HPV vaccine in 2019 and 2024. The data show an increase in vaccination rates across all age groups during this period. Overall, the vaccination rate rose from 6% (42/662) in 2019 to 30% (108/602) in 2024. Among girls aged 12–14 and 15–17, the rates increased modestly, from 1% to 2% and 3% to 6%, respectively. The most significant increase was observed in the 18+ age group, where the vaccination rate increased from 2%

Table 1. Demographic characteristics of parents recruited in 2019, parents recruited in 2024, and parents with daughters 9–17 years old in 2024, Shanghai, China.

	2019 N = 1037	2024 All parents N = 1450	2024 parents of daughter 9–17 years old N = 449
Age of parent			
18–29	73 (11%, 7%–15%)	45 (11%, 8%–14%)	4 (3%, <0.5 %– 6%)
30–39	595 (62%, 57%–68%)	801 (62%, 59%–65%)	205 (57%, 53%–62%)
40+	347 (27%, 23%–31%)	604 (27%, 25%–29%)	240 (39%, 35%–44%)
Relation to child			
Mother	818 (83%, 79%–88%)	1186 (83%, 81%–85%)	373 (83%, 79%–87%)
Father	203 (17%, 13%–21%)	249 (16%, 14%–18%)	71 (16%, 12%–20%)
Grandparents/others		15 (1%, <0.5%–1%)	5 (1%, <0.5%–1%)
Parents' education			
High school or less	129 (12%, 8%–16%)	78 (5%, 4%–7%)	29 (8%, 5%–11%)
Vocational degree/certificate	389 (35%, 30%–40%)	613 (44%, 41%–47%)	185 (46%, 40%–51%)
College or above	498 (53%, 47%–58z5)	759 (50%, 47%–53%)	235 (47%, 42%–52%)
Monthly household income			
<9,999yuan	336 (33%, 27%–38%)	334 (25%, 23%–28%)	110 (27%, 23%–32%)
10,000–19,999 yuan	392 (37%, 32%–43%)	614 (40%, 37%–43%)	188 (40%, 35%–45%)
≥20,000 yuan	295 (30%, 25%–35%)	502 (34%, 32%–37%)	151 (33%, 28%–37%)
Huji			
Local (SH)	706 (65%, 60%–71%)	1069 (65%, 62%–68%)	342 (68%, 63%–73%)
Non-local	321 (35%, 29%–40%)	381 (35%, 32%–38%)	107 (32%, 27%–37%)
Hukou (Urbanicity)			
City	746 (72%, 67%–77%)	1174 (72%, 69%–75%)	367 (73%, 68%–78%)
Rural	280 (28%, 23%–33%)	276 (28%, 25%–31%)	82 (27%, 22%–32%)
District			
Urban	205 (30%, 25%–34%)	375 (30%, 27%–33%)	118 (27%, 23%–32%)
Suburban	832 (70%, 66%–75%)	1075 (70%, 67%–73%)	331 (73%, 68%–77%)
General Vaccine hesitancy			
Hesitant overall	228 (18%, 14%–21%)	163 (11%, 10%–13%)	40 (9%, 6%–12%)
Hesitant, general attitudes subdomain	133 (9%, 7%–11%)	116 (8%, 6%–10%)	25 (6%, 4%–8%)
Hesitant, safety and efficacy subdomain	729 (68%, 63%–73%)	279 (20%, 18%–22%)	77 (17%, 13%–20%)
Hesitant, behavior subdomain	96 (10%, 6%–13%)	392 (27%, 24%–29%)	127 (27%, 22%–31%)
Vaccine hesitant to HPV vaccine			
Hesitant	/	410 (28%, 25%–31%)	124 (26%, 22%–31%)
Non-hesitant	/	1040 (72%, 69%–75%)	325 (74%, 69%–78%)
Knowledge of HPV			
High	504 (51%, 45%–56%)	977 (67%, 64%–70%)	297 (63%, 59%–68%)
Low	533 (49%, 44%–55%)	473 (33%, 30%–36%)	152 (37%, 32%–41%)

Table 2. Cross-tabulations between demographic characteristics and vaccine hesitancy in Shanghai, China, 2024.

	Hesitant Overall N = 1450	Vaccine hesitant to HPV vaccine (daughter 9–17) N = 449
Overall	163	124
Age of parent		
18–29	5 (12%, 2%–23%)	1 (21%, <0.5%–59%)
30–39	81 (11%, 8%–13%)	45 (22%, 17%–28%)
40+	77 (13%, 10%–16%)	78 (33%, 27%–39%)
Relation to child		
Mother	126 (11%, 9%–13%)	90 (23%, 19%–28%)
Father	37 (14%, 10%–19%)	32 (42%, 29%–54%)
Grandparents/others	0	2 (22%, <0.5%–83%)
Parents' education		
High school or less	14 (15%, 7%–23%)	9 (29%, 12%–46%)
Vocational degree/certificate	70 (11%, 8%–13%)	56 (29%, 22%–36%)
College or above	79 (12%, 9%–15%)	59 (24%, 18%–29%)
Monthly household income		
<9,999yuan	50 (16%, 11%–20%)	38 (33%, 24%–42%)
10,000–19,999 yuan	56 (9%, 6%–11%)	49 (24%, 18%–31%)
≥20,000 yuan	57 (12%, 9%–15%)	37 (24%, 17%–31%)
Huji		
Local (SH)	107 (10%, 8%–12%)	92 (26%, 21%–31%)
Non-local	56 (15%, 11%–19%)	32 (28%, 20%–37%)
Hukou (Urbanicity)		
City	124 (11%, 9%–13%)	99 (27%, 22%–32%)
Rural	39 (12%, 8%–16%)	25 (26%, 16%–35%)
District		
Urban	45 (14%, 10%–19%)	31 (26%, 18%–35%)
Suburban	118 (10%, 8%–12%)	93 (26%, 21%–32%)
Knowledge of HPV		
High	89 (9%, 7%–12%)	71 (24%, 18%–29%)
Low	74 (16%, 12%–19%)	53 (31%, 24%–39%)

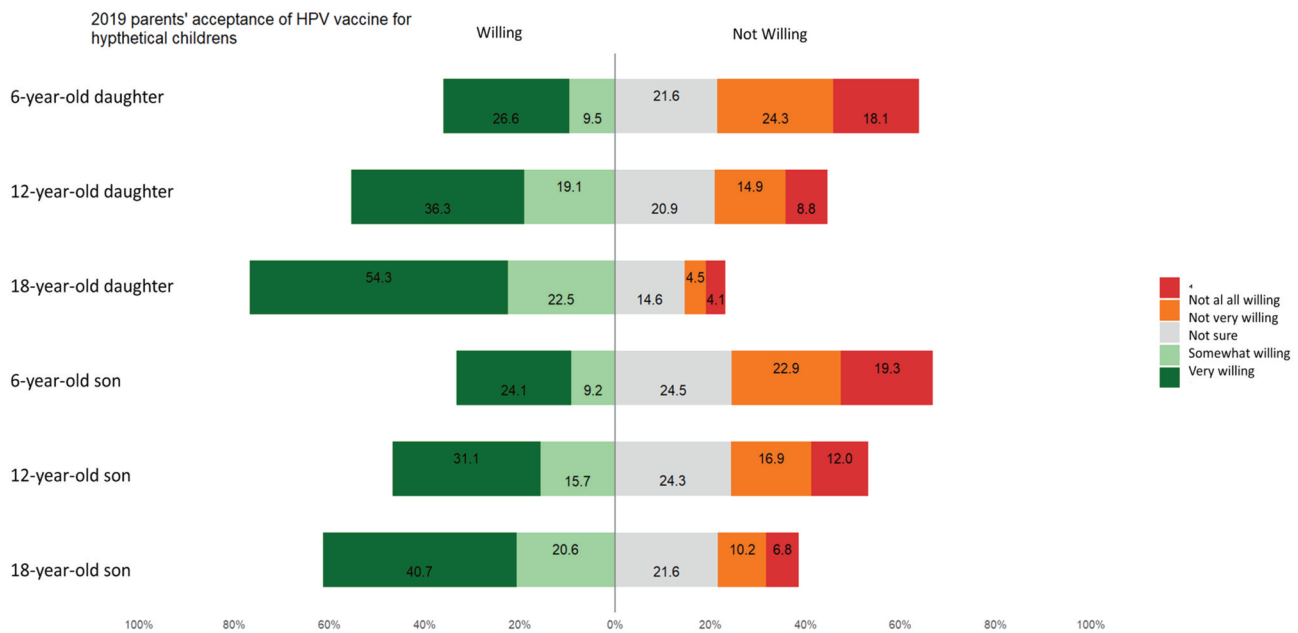


Figure 1. Parents' acceptance of HPV vaccine for their hypothetical child by age and gender in 2019.

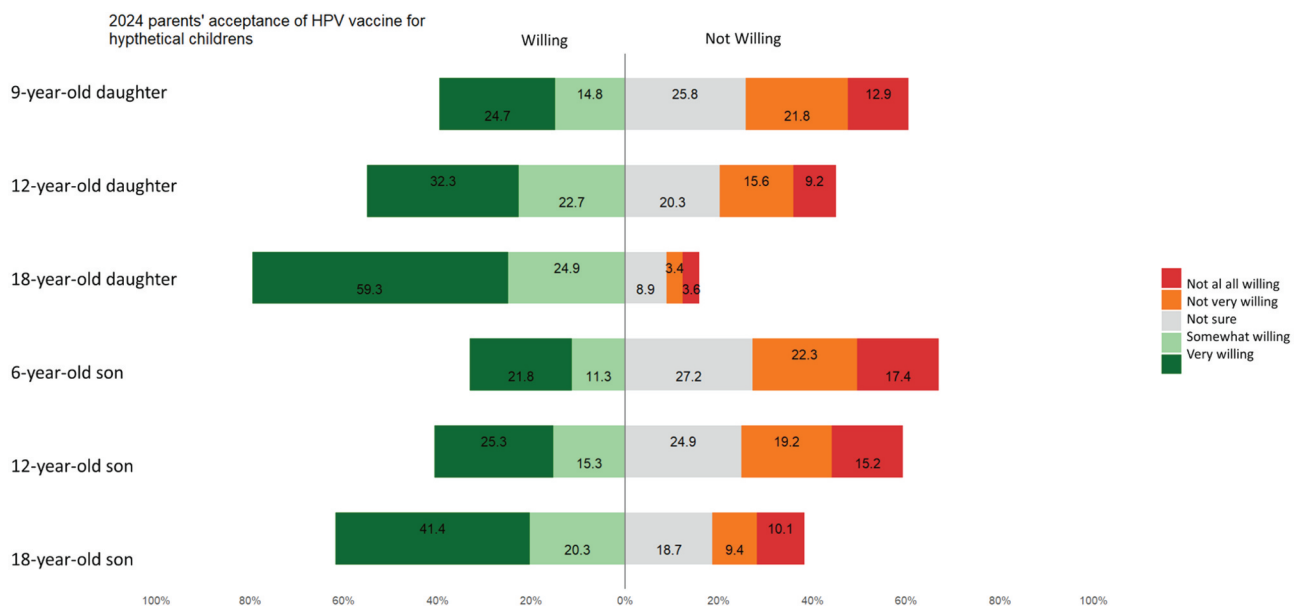


Figure 2. Parents' acceptance of HPV vaccine for their hypothetical child by age and gender in 2019.

(10/662) in 2019 to 16% (96/602) in 2024. These results highlight that older adolescents remain the primary population receiving the HPV vaccine.

Table 3 shows the associations between general vaccine hesitancy, HPV-specific vaccine hesitancy, and parents' decisions to vaccinate their daughters aged 9–17 years. Parents with general vaccine hesitancy were 94% less likely to vaccinate their daughters with the HPV vaccine (OR = 0.06, 95% CI: 0.00–0.89). Similarly, parents hesitant to the HPV vaccine specifically were 75% less likely to vaccinate their daughters (OR = 0.25, 95% CI: 0.11–0.56). Additionally, parents were more likely to vaccinate older daughters. Compared to girls aged 9–11, parents of girls aged 12–14 and 15–17 were approximately 4 times more likely to

vaccinate their daughters, regardless of vaccine hesitancy type (e.g., 12–14 vs. 9–11: OR = 4.49, 95% CI: 2.02–10.02 for general hesitancy). Similar age-related patterns were observed in the safety, efficacy, and behavior subdomains, as shown in Appendix Table A1. However, the general attitude subdomain was inestimable and is therefore not included in the table.

Population Attributable Fraction (PAF)

Table 4 summarizes the PAF results. Among the study population, 10% of HPV non-vaccination could be attributed to general vaccine hesitancy (PAF = 10.07%, 97.5% CI: –2.17%–14.36%), while 24% could be attributed to HPV-specific

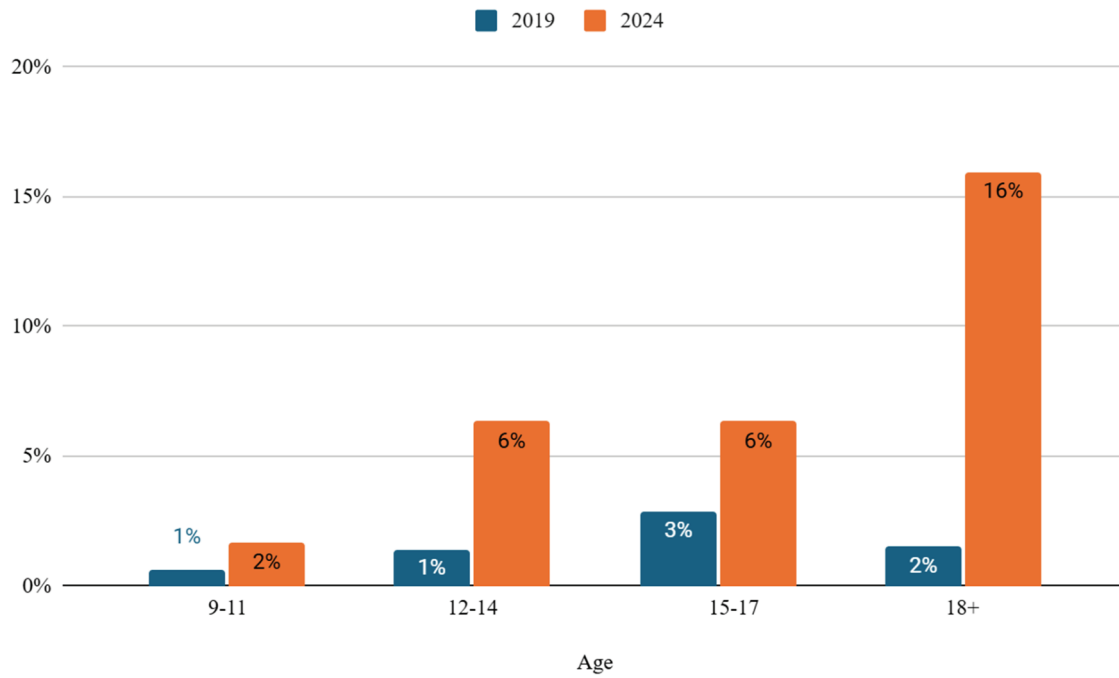


Figure 3. Percentage of HPV vaccination rate by age among girls who are eligible(9+) in 2019 ($N = 662$) and 2024 ($N = 602$).

Table 3. Multivariable logistic regression model of vaccine hesitancy, HPV-specific vaccine hesitancy, and HPV vaccination among girls 9–17 years old in Shanghai, China, 2024.

	HPV vaccination	<i>p</i> value	HPV vaccination	<i>p</i> value
Hesitant Overall	0.06 (0.00, 0.89)	0.0013		
Vaccine hesitant to HPV vaccine	–	–	0.25 (0.11, 0.56)	0.0002
Demographic Variables				
Age of parent		0.30		0.10
18–29	1.55 (0.29, 8.27)		2.20 (0.38, 12.68)	
30–39	0.65 (0.33, 1.27)		0.56 (0.28, 1.11)	
40+	ref		ref	
Relation to child		0.58		0.47
Mother	0.26 (0.02, 3.35)		0.21 (0.01, 3.58)	
Father	0.31 (0.02, 4.26)		0.29 (0.02, 5.19)	
Grandparents/others	ref		ref	
Parents' education		0.74		0.54
High school or less	1.63 (0.48, 5.48)		2.01 (0.59, 6.86)	
Vocational degree/certificate	1.11 (0.58, 2.12)		1.16 (0.61, 2.20)	
College or above	ref		ref	
Monthly household income		0.43		0.32
<9,999yuan	1.29 (0.60, 2.80)		1.22 (0.56, 2.62)	
10,000–19,999 yuan	0.80 (0.41, 1.59)		0.71 (0.36, 1.40)	
≥20,000 yuan	ref		ref	
Huji		0.91		0.90
Local (SH)	0.95 (0.43, 2.13)		1.06 (0.47, 2.36)	
Non-local	ref		ref	
Hukou (Urbanicity)		0.87		0.49
City	0.93 (0.36, 2.37)		0.72 (0.29, 1.82)	
Rural	ref		ref	
District		0.84		0.97
Urban	0.93 (0.47, 1.84)		1.01 (0.52, 1.99)	
Suburban	ref		ref	
Knowledge of HPV		0.05		0.06
High	ref		ref	
Low	1.93 (0.98, 3.80)		1.88 (0.95, 3.70)	
Child's age		0.0001		0.0002
9–11	ref		ref	
12–14	4.49 (2.02, 10.02)		4.09 (1.84, 9.11)	
15–17	4.43 (1.80, 10.90)		4.70 (1.91, 11.58)	

Table 4. Population-attributable fractions (PAF) of vaccine hesitancy factors contributing to HPV vaccination uptake among girls aged 9–17 in Shanghai, China.

Predictor variable	Proportion hesitant among vaccinated (proportion, 97.5%CI)	Proportion hesitant among unvaccinated (proportion, 97.5%CI)	Prevalence ratio of unvaccinated (97.5%CI)	PAF (LL, UL)
Hesitant Overall	1.16 (0, 3.77)	10.74 (7.08, 14.40)	16.01 (0.77, 334.69)	10.07% (–2.17%, 14.36%)
Vaccine hesitant to HPV vaccine	11.63 (3.85, 19.41)	31.41 (25.92, 36.89)	4.04 (1.60, 10.20)	23.63% (9.72%, 33.27%)

models adjusted for household monthly income, parents' education, parental age, relationship to children, parent' HPV knowledge, children's age, Huiji (local-non-local), hukou (urbanicity), district (central-suburban district).

vaccine hesitancy (PAF = 23.63%, 97.5% CI: 9.72%–33.27%). Less than 4% of non-vaccination cases were attributable to any vaccine hesitancy subdomains (Appendix Table A3).

Discussion

In our serial cross-sectional study conducted in 2019 and 2024 in Shanghai, China, a significant increase in HPV vaccination among girls was observed over the 5 years, climbing from 6% to 30%, particularly among those aged 18 and older. This increase aligns with parents' acceptance of the vaccine when asked in the context of hypothetical children. However, while overall acceptance improved, acceptance for the younger age group decreased. That means the increased vaccination rate mostly come from the older age group children. Our analysis further confirmed that older age is significantly associated with higher HPV vaccination rates. We also found that HPV-specific vaccine hesitancy accounted for about 24% of HPV non-vaccination, suggesting that eliminating HPV-specific hesitancy could substantially reduce non-vaccination rates.

Since initial HPV vaccine approval in 2016, China has made notable progress in increasing vaccination coverage through policy changes, domestic vaccine production, and public health initiatives. These efforts have addressed key barriers such as cost, awareness, and accessibility to better align with global cervical cancer elimination goals. Although parental knowledge of HPV has improved from 51% to 67% in our study, HPV-specific vaccine hesitancy remains high at 28%, indicating that knowledge alone may not be sufficient to shift attitudes and actions. Additional challenges such as uneven vaccine distribution, competing health priorities, and persistent misinformation continue to hinder the vaccine uptake, highlighting the need for targeted behavioral and communication strategies along with the structural improvements.¹⁶

Although overall HPV vaccination coverage has increased, most recipients are young adults aged 18 and older, while vaccination coverage for girls aged 9–14 remains very low. This finding is consistent with other studies across China. A study in Shenzhen, a developed southern city, reported that 23% of women aged 18–26 received at least one dose of the HPV vaccine, while coverage for girls aged 9–17 was less than 5%.¹⁷ National data also indicates that coverage among the WHO target age group of 9–14 years was only 4%,⁸ far below the rates in Japan (26%), South Africa (49%), and the United States (80%), and significantly short of the 90% target set by WHO.¹⁸

Our findings indicate that the lower coverage for younger age groups may result from vaccine supply constraints, limited-service accessibility, high vaccine costs, and insufficient awareness about

the optimal timing for vaccination.¹⁹ In the past, HPV vaccines in China were primarily recommended for adult women due to limited supply and age restrictions.¹⁹ For instance, the bivalent HPV vaccine was restricted to females aged 9–25 before 2018, the quadrivalent vaccine was only available for ages 20–40 years before 2019. The nonavalent vaccine (protect against nine different HPV serotypes) was limited to ages 16–26 years before 2021.⁸ In 2022, vaccine eligibility was expanded to include women aged 9–45 years. The quadrivalent vaccine was approved for men 9–26 in January 2025. However, a lack of clear understanding about the different vaccine types among the public continues to hinder informed decision-making.¹⁶ An over-preference for the nonavalent vaccine has created supply shortages, making it difficult for individuals to access to the vaccine.²⁰ A study involving over 4,000 eligible parents in China showed that among those intending to vaccinate their daughters, 62% attempted to schedule a vaccination appointment, but only 16% succeeded.²⁰ These persistent challenges have led some women to pay extra for vaccination at private clinics or turn to scalpers, who resell vaccine slots at inflated prices.²¹ This “scalping” phenomenon has contributed to frustration and distrust, which may translate into hesitancy or even abandoning plans for vaccination. Thus, these structural constraints not only limit physical access but also reinforce psychological barriers to vaccine acceptance. Besides increasing the vaccine supply, objective education about vaccine choices, consistent updates on government funding policies, and sustained improvements in public health services are needed to address these challenges.

Our PAF analysis showed that general vaccine hesitancy accounted for 10% of HPV non-vaccination, while HPV-specific vaccine hesitancy contributed 24%. General vaccine hesitancy involves issues such as medical mistrust, exposure to misinformation about vaccines, and concerns about safety and efficacy.²² HPV-specific hesitancy, however, includes unique barriers such as stigma associated with HPV as an STI, low awareness of the link between HPV and cancer, and cultural factors such as gender norms, taboos around sex and STIs and religious beliefs.^{12,13} These findings align with studies conducted in other countries, such as the United States, where parental hesitancy for the HPV vaccine was found to be about four times higher than for other childhood vaccines.²³ HPV-specific concerns include fears of vaccine novelty,²³ potential serious side effects (e.g., vaginal bleeding),²⁴ myths about a link to infertility, and worries that vaccination may encourage sexual activity (so called, sexual disinhibition).²⁵ Our earlier study demonstrated that framing the HPV vaccine as a cancer-prevention tool and providing information about its role in preventing sexually transmitted infections improved parents' acceptance.¹⁵ These results emphasize the importance of addressing HPV-specific vaccine hesitancy through

tailored interventions. Future research should focus on understanding specific barriers in greater detail and developing messaging that reframes the HPV vaccine as a tool for preventing multiple types of cancer.

To improve HPV vaccination uptake among girls aged 9–14, several actionable strategies can be implemented in response to our findings. First, school-based vaccination programs have been effective in many countries, as they reach adolescents in a structured and accessible setting, thus reducing logistical barriers for families. A nationwide study in France evaluating various interventions to improve HPV vaccination rate found that school-based programs were the only approach that consistently and significantly increased vaccination coverage, even after the trial ended.²⁶ Second, parental education sessions held during school meetings delivered by healthcare providers using tailored messaging are essential for addressing misinformation about HPV and the vaccine. This can help with clarifying differences between vaccine types, potential side effects and empowering parents to make informed decisions. Lastly, up-to-date information about reimbursement policies and vaccine availability should be communicated to parents regularly to reduce cost-related hesitancy and encourage timely vaccination.

There are strengths and limitations to our study. The serial cross-sectional design allowed us to observe changes in parental attitudes over time and assess the impact of new policies to promote the HPV vaccine. Matching the 2019 and 2024 datasets on key socioeconomic variables enhanced comparability. Additionally, the PAF analysis quantified the contribution of general and HPV-specific vaccine hesitancy to non-vaccination, helping prioritize strategies for promoting vaccination. However, as a cross-sectional study, it is subject to unmeasured confounding and imprecise measurements, limiting causal inferences. The PAF estimates are approximations and may not fully capture the complex associations between non-vaccination and hesitancy factors. Residual confounding may also persist despite matching efforts.

Conclusion

In conclusion, while HPV vaccination rates in China have improved, disparities remain, particularly among younger age groups. Addressing HPV-specific vaccine hesitancy, logistical barriers, and knowledge gaps through tailored interventions and public education is essential to meet global vaccination targets and reduce the burden of cervical cancer in China.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This work did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Funding

This study was supported by the National Institute Of Allergy And Infectious Diseases of the National Institutes of Health under Award Number K01AI137123. ZH, HY, and JR were supported Key Discipline-

Infectious Diseases (No. GWVI-11.1-01) of Three-year Action Program of Shanghai Municipality for Strengthening the Construction of Public Health System (2023–2025).

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Appendix

Table A1. Association between demographic characteristics and vaccine hesitancy subdomains in Shanghai, China, 2024.

	Hesitant, general attitudes subdomain N = 1450	Hesitant, safety and efficacy subdomain N = 1450	Hesitant, behavior subdomain N = 1450
Overall	116 (8%, 6%-10%)	279 (20%, 18%-22%)	392 (27%, 24%-29%)
Age of parent			
18–29	4 (10%, 0–19%)	9 (21%, 9%-34%)	12 (29%, 15%-43%)
30–39	54 (7%, 5%-9%)	154 (20%, 17%-23%)	207 (25%, 22%-28%)
40+	58 (10%, 7%-12%)	116 (20%, 17%-23%)	173 (28%, 25%-32%)
Relation to child			
Mother	89 (8%, 6%-10%)	219 (20%, 17%-22%)	329 (27%, 24%-30%)
Father	27 (10%, 6%-13%)	57 (22%, 16%-27%)	59 (22%, 17%-27%)
Grandparents/others	0	3 (16%, 0%-33%)	4 (31%, 6%-56%)
Parents' education			
High school or less	10 (11%, 4%-17%)	23 (23%, 14%-32%)	15 (21%, 11%-31%)
Vocational degree/certificate	52 (7%, 5%-10%)	111 (19%, 15%-23%)	168 (26%, 22%-30%)
College or above	54 (8%, 6%-11%)	145 (20%, 17%-24%)	209 (27%, 24%-31%)
Monthly household income			
<9,999yuan	35 (12%, 7%-16%)	90 (27%, 22%-33%)	94 (28%, 22%-33%)
10,000–19,999 yuan	42 (6%, 4%-8%)	92 (15%, 12%-18%)	159 (26%, 22%-30%)
≥20,000 yuan	39 (8%, 5%-10%)	97 (20%, 16%-25%)	139 (27%, 22%-31%)
Huji			
Local (SH)	80 (7%, 6%-9%)	186 (18%, 15%-20%)	300 (27%, 24%-30%)
Non-local	36 (9%, 6%-12%)	93 (24%, 19%-29%)	92 (26%, 21%-31%)
Hukou (Urbanicity)			
City	94 (8%, 6%-10%)	206 (18%, 16%-20%)	334 (28%, 25%-31%)
Rural	22 (7%, 4%-11%)	73 (25%, 19%-31%)	58 (22%, 16%-28%)
District			
Urban	36 (11%, 7%-15%)	70 (21%, 16%-26%)	107 (28%, 23%-33%)
Suburban	80 (7%, 5%-8%)	209 (20%, 17%-22%)	285 (26%, 23%-29%)
Knowledge of HPV			
High	59 (6%, 4%-8%)	169 (18%, 15%-21%)	262 (26%, 23%-29%)
Low	57 (12%, 8%-15%)	110 (24%, 20%-29%)	130 (28%, 23%-33%)

Table A2. Multivariable Poisson regression model of vaccine hesitancy subdomains and HPV vaccination among girls 9–17 in Shanghai, China, 2024.

	HPV vaccination	P value	HPV vaccination	P value
Hesitant, safety, and efficacy subdomain	0.80 (0.37, 1.76)	0.58		
Hesitant, behavior subdomain			0.90 (0.48, 1.70)	0.75
Demographic Variables				
Age of parent		0.23		0.23
18–29	1.85 (0.35, 9.88)		1.91 (0.36, 10.15)	
30–39	0.64 (0.33, 1.25)		0.65 (0.33, 1.27)	
40+	ref		ref	
Relation to child		0.50		0.51
Mother	0.22 (0.02, 2.85)		0.22 (0.02, 2.87)	
Father	0.27 (0.02, 3.74)		0.26 (0.02, 3.71)	
Grandparents/others	ref		ref	
Parents' education		0.58		0.57
High school or less	1.85 (0.56, 6.09)		1.88 (0.57, 6.17)	
Vocational degree/certificate	1.04 (0.55, 1.98)		1.05 (0.56, 2.00)	
College or above	ref		ref	
Monthly household income		0.44		0.48
<9,999yuan	1.12 (0.52, 2.38)		1.09 (0.51, 2.30)	
10,000–19,999 yuan	0.73 (0.37, 1.43)		0.73 (0.37, 1.43)	
≥20,000 yuan	ref		ref	
Huji		0.91		0.91
Local (SH)	0.96 (0.43, 2.11)		0.95 (0.43, 2.11)	
Non-local	ref		ref	
Hukou (Urbanicity)		0.72		0.70
City	0.85 (0.34, 2.10)		0.84 (0.34, 2.06)	
Rural	ref		ref	
District		0.92		0.95
Urban	1.03 (0.53, 2.00)		1.02 (0.53, 1.98)	
Suburban	ref		ref	
Knowledge of HPV		0.03		0.03
High	ref		ref	
Low	2.02 (1.03, 3.95)		2.04 (1.04, 3.99)	
Children's age		0.0002		0.0002
9–11	ref		ref	
12–14	4.19 (1.89, 9.25)		4.18 (1.89, 9.24)	
15–17	4.31 (1.76, 10.51)		4.43 (1.79, 10.94)	

Table A3. Population-attributable fractions (PAF) of vaccine hesitancy factors contributing to HPV vaccination uptake among girls aged 9–17 in Shanghai, China.

	Proportion hesitant among vaccinated (proportion,97.5%CI)	Proportion hesitant among unvaccinated (proportion,97.5%CI)	Prevalence ratio of unvaccinated (97.5%CI)	
Hesitant, safety, and efficacy subdomain	13.95 (5.54, 22.37)	17.91 (13.38, 22.44)	1.04 (0.90, 1.19)	3.50% (–13.00%, 15.08%)
Hesitant, behavior subdomain	27.91 (17.02, 38.80)	28.37 (23.05, 33.70)	1.02 (0.91, 1.13)	2.75% (–19.80%, 18.92%)